



华南理工大学
South China University of Technology

操作系统实验报告

郑重声明：

- 1、实验手册中的所有实验均有本人独立编码、调试和测试。
- 2、实验手册中给出的实验数据和结果完全由本人所完成的程序给出。
- 3、本人了解：不按照前两条要求所完成的实验报告已经构成了抄袭或造假行为，本人将承担相应的不良后果。

姓名：_____（签名）

学号：_____

班级：_____22 计算机科学与技术 1 班

提交日期：_____2025/06/08

成绩：_____

一、实验目的

1. 了解使用 Linux 操作系统，熟悉 Linux 桌面环境（GNOME/KDE），掌握基本操作
2. 学习常用的 Shell 命令的使用和 Shell 脚本编写
3. 学会使用 vi 编辑器进行文档编辑
4. 学习 Linux 系统文件与目录管理
5. 学习 Linux 系统监视与进程管理
6. 学习 Linux 下程序编辑，编译和执行

二、实验内容

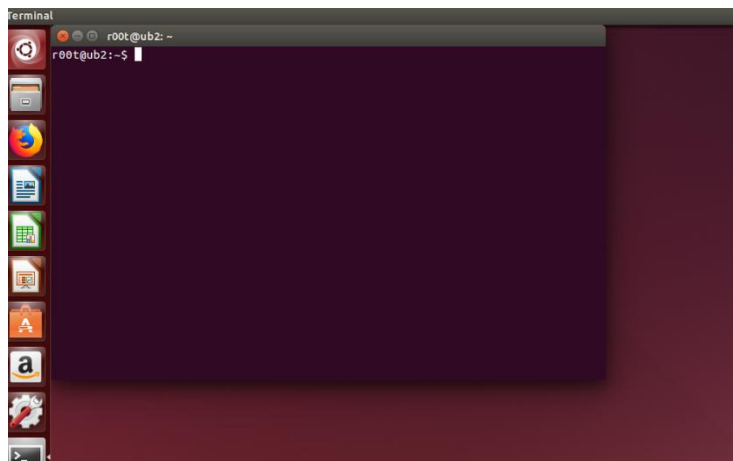
1. 认识 Linux 桌面环境（GNOME）
2. 常用 Shell 命令的使用
3. Vi 编辑器的使用
4. Shell 脚本编写
5. 学习 Linux 文件系统与目录管理

三、实验环境

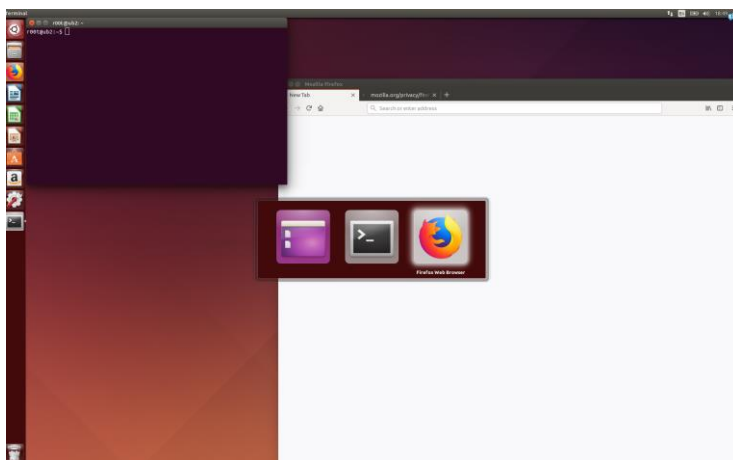
Ubuntu14.04

四、实验步骤（附上截图）

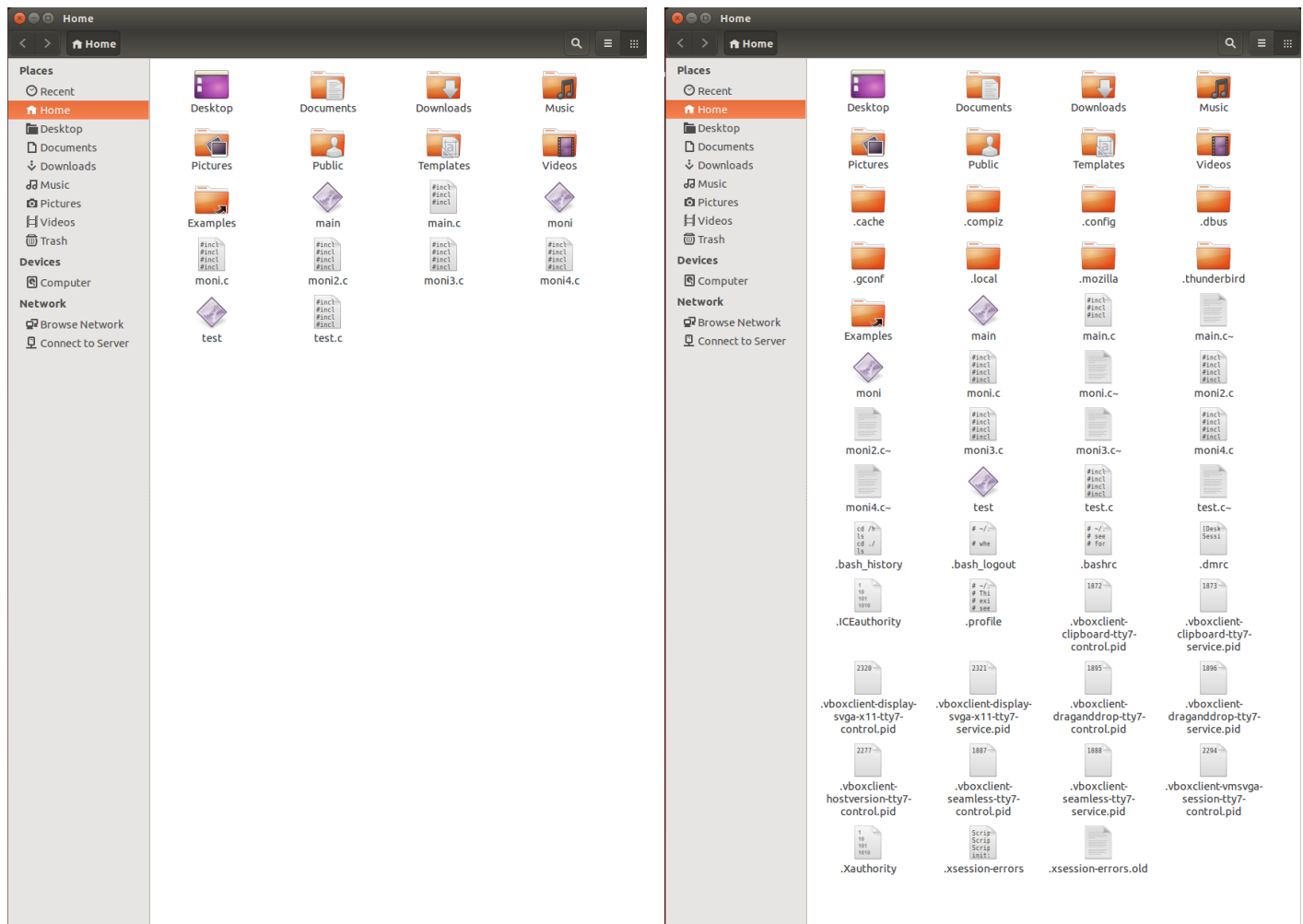
1. 认识 Linux 桌面环境（GNOME）
 - 1.1 了解常用快捷键



Ctrl+Alt+T 打开终端

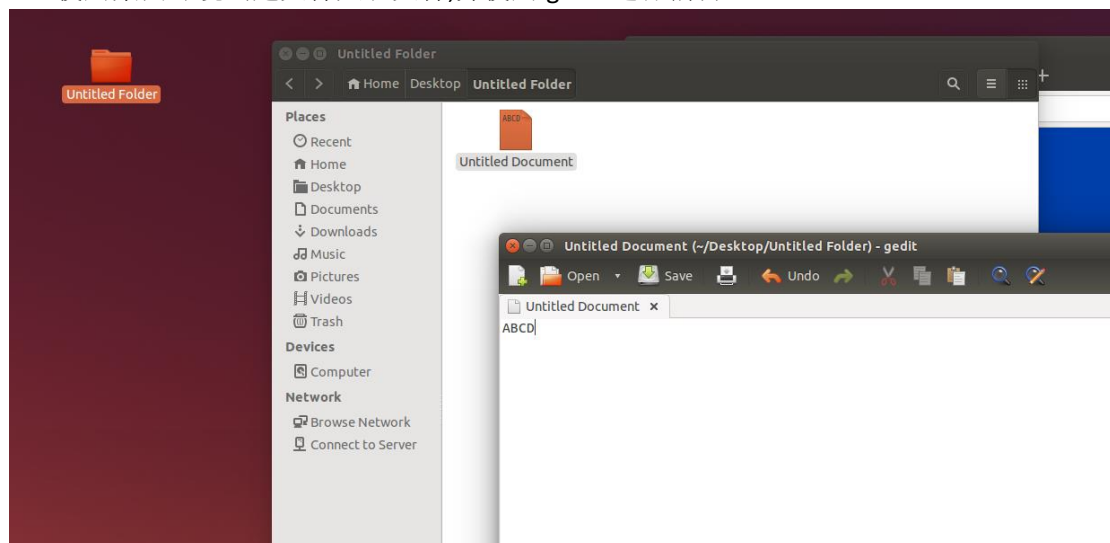


Tab+Alt 窗口切换



Ctrl+H 显示隐藏文件(左: 执行前, 右: 执行后)

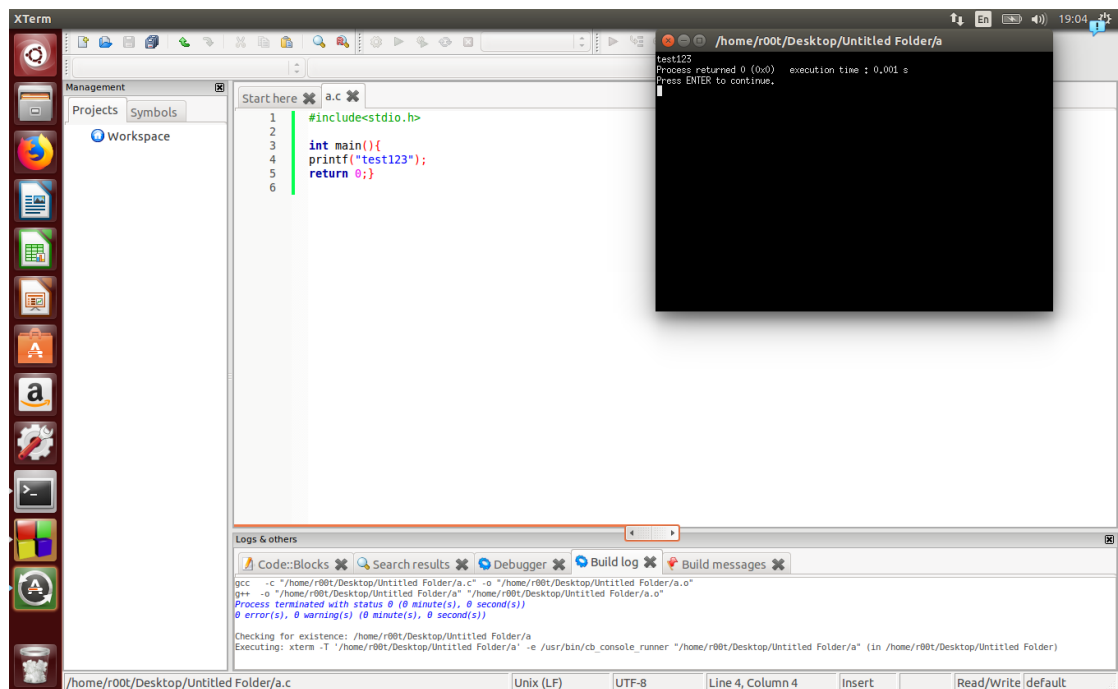
1.2 使用桌面环境创建文件夹和文件,并使用 gedit 进行编辑



1.3 使用 FireFox 浏览网页



1.4 下载使用 Eclipse 等 IDE



2. 常用 Shell 命令的使用

(1) 在 root 下创建一个目录 mydir

```
root@ub2:/home/r00t# cd /root
root@ub2:~/# mkdir mydir
root@ub2:~/# ls
mydir  vboxpostinstall.sh
```

(2) 复制文件/etc/inittab 到 mydir 目录下

```
root@ub2:~# cp /etc/inittab mydir/
cp: cannot stat '/etc/inittab': No such file or directory
root@ub2:~# cp /etc/passwd mydir/
root@ub2:~# ls /mydir
ls: cannot access /mydir: No such file or directory
root@ub2:~#
root@ub2:~# ls ./mydir
passwd
root@ub2:~#
```

此版本系统中, /etc/inittab 不存在, 改为复制/etc/passwd 测试

(3) 在 mydir 目录下创建文件 mydoc

```
root@ub2:~# cd ./mydir
root@ub2:~/mydir# touch mydoc
root@ub2:~/mydir# ls
mydoc passwd
root@ub2:~/mydir#
```

(4) 删除 mydir 目录

```
root@ub2:~# cd ./mydir
root@ub2:~/mydir# cd ../
root@ub2:~# rm -r ./mydir
root@ub2:~#
```

(5) 统计文件/etc/inittab 的字符数和行数

```
root@ub2:~#
root@ub2:~# wc inittab
83 452 3067 inittab
root@ub2:~#
```

此版本系统中, /etc/inittab 不存在, 我从

<https://github.com/brgl/busybox/blob/master/examples/inittab> 下载 inittab 文件到本机后再进行统计操作

(6) 用列表方式显示 etc 目录下的所有文件

```
root@ub2:~# ls -l /etc
total 1132
drwxr-xr-x 3 root root 4096 Mar 5 2019 acpi
-rw-r--r-- 1 root root 2981 Mar 5 2019 adduser.conf
drwxr-xr-x 2 root root 12288 May 28 19:32 alternatives
-rw-r--r-- 1 root root 401 Feb 20 2014 anacrontab
-rw-r--r-- 1 root root 112 Jan 10 2014 apg.conf
drwxr-xr-x 6 root root 4096 Mar 5 2019 apn
drwxr-xr-x 3 root root 4096 Mar 5 2019 apparmor
drwxr-xr-x 9 root root 4096 Mar 5 2019 apparmor.d
drwxr-xr-x 5 root root 4096 Mar 5 2019 appport
drwxr-xr-x 6 root root 4096 Jun 4 18:59 apt
drwxr-xr-x 3 root root 4096 Mar 5 2019 aptdaemon
drwxr-xr-x 2 root root 4096 Mar 5 2019 at-spi2
drwxr-xr-x 3 root root 4096 Mar 5 2019 avahi
-rw-r--r-- 1 root root 2177 Apr 9 2014 bash.bashrc
-rw-r--r-- 1 root root 45 Mar 23 2014 bash_completion
drwxr-xr-x 2 root root 4096 May 28 19:34 bash_completion.d
-rw-r--r-- 1 root root 356 Jan 2 2012 bindresvport.blacklist
-rw-r--r-- 1 root root 321 Apr 17 2014 blkid.conf
lrwxrwxrwx 1 root root 15 May 26 14:52 blkid.tab -> /dev/.blkid.tab
drwxr-xr-x 2 root root 4096 Mar 5 2019 bluetooth
-rw-r--r-- 1 root root 33 Mar 5 2019 brlapi.key
drwxr-xr-x 2 root root 12288 Mar 5 2019 brltty
-rw-r--r-- 1 root root 22478 Apr 1 2014 brltty.conf
drwxr-xr-x 3 root root 4096 Mar 5 2019 ca-certificates
-rw-r--r-- 1 root root 6488 Mar 5 2019 ca-certificates.conf
drwxr-xr-x 2 root root 4096 Mar 5 2019 calendar
drwxr-xr-x 2 root root 4096 Mar 5 2019 chatscripts
-rw-r--r-- 1 root root 1332 Jan 16 2014 colord.conf
drwxr-xr-x 2 root root 4096 Mar 5 2019 compizconfig
drwxr-xr-x 2 root root 4096 Mar 5 2019 console-setup
drwxr-xr-x 2 root root 4096 Mar 5 2019 cracklib
drwxr-xr-x 2 root root 4096 Mar 5 2019 cron.d
drwxr-xr-x 2 root root 4096 Mar 5 2019 cron.daily
drwxr-xr-x 2 root root 4096 Mar 5 2019 cron.hourly
drwxr-xr-x 2 root root 4096 Mar 5 2019 cron.monthly
-rw-r--r-- 1 root root 722 Feb 9 2013 crontab
drwxr-xr-x 2 root root 4096 Mar 5 2019 cron.weekly
drwxr-xr-x 5 root lp 4096 Jun 4 19:02 cups
drwxr-xr-x 2 root root 4096 Mar 5 2019 cupshelpers
drwxr-xr-x 4 root root 4096 Mar 5 2019 dbus-1
drwxr-xr-x 4 root root 4096 Mar 5 2019 dconf
-rw-r--r-- 1 root root 2969 Feb 23 2014 debconf.conf
-rw-r--r-- 1 root root 11 Feb 20 2014 debian_version
drwxr-xr-x 2 root root 4096 May 28 19:32 default
-rw-r--r-- 1 root root 604 Nov 7 2013 deluser.conf
drwxr-xr-x 2 root root 4096 May 26 15:07 depmod.d
drwxr-xr-x 4 root root 4096 Mar 5 2019 dhcp
drwxr-xr-x 2 root root 4096 Mar 5 2019 dictionaries-common
drwxr-xr-x 2 root root 4096 Mar 5 2019 desktop.d
```

(7) 用 cat、more、less、head、tail 查看文件/etc/inittab

```
root@ub2:~# cat inittab
# /etc/inittab init(8) configuration for BusyBox
#
# Copyright (C) 1999-2004 by Erik Andersen <andersen@codepoet.org>
#
# Note, BusyBox init doesn't support runlevels. The runlevels field is
# completely ignored by BusyBox init. If you want runlevels, use sysvinit.
#
# Format for each entry: <id>:<runlevels>:<action>:<process>
#
# <id>: WARNING: This field has a non-traditional meaning for BusyBox init!
#
# The id field is used by BusyBox init to specify the controlling tty for
# the specified process to run on. The contents of this field are
# appended to "/dev/" and used as-is. There is no need for this field to
# be unique, although if it isn't you may have strange results. If this
# field is left blank, then the init's stdin/out will be used.
#
# <runlevels>: The runlevels field is completely ignored.
#
# <action>: Valid actions include: sysinit, respawn, askfirst, wait, once,
# restart, ctrlaltdel, and shutdown.
#
# Note: askfirst acts just like respawn, but before running the specified
# process it displays the line "Please press Enter to activate this
# console." and then waits for the user to press enter before starting
# the specified process.
#
# Note: unrecognized actions (like initdefault) will cause init to emit
# an error message, and then go along with its business.
#
# <process>: Specifies the process to be executed and it's command line.
#
# Note: BusyBox init works just fine without an inittab. If no inittab is
# found, it has the following default behavior:
#
# ::sysinit:/etc/init.d/rcS
# ::askfirst:/bin/sh
# ::ctrlaltdel:/sbin/reboot
# ::shutdown:/sbin/swapoff -a
# ::shutdown:/bin/umount -a -r
# ::restart:/sbin/init
# tty2::askfirst:/bin/sh
# tty3::askfirst:/bin/sh
# tty4::askfirst:/bin/sh
#
# Boot-time system configuration/initialization script.
# This is run first except when booting in single-user mode.
#
# ::sysinit:/etc/init.d/rcS
#
# /bin/sh invocations on selected ttys
#
# Note below that we prefix the shell commands with a "-" to indicate to the
# shell that it is supposed to be a login shell. Normally this is handled by
# login, but since we are bypassing login in this case, BusyBox lets you do
# this yourself...
#
# Start an "askfirst" shell on the console (whatever that may be)
# ::askfirst:-/bin/sh
# Start an "askfirst" shell on /dev/tty2-4
# tty2::askfirst:-/bin/sh
# tty3::askfirst:-/bin/sh
# tty4::askfirst:-/bin/sh
```

```
root@ub2:~# cat /etc/inittab
# /etc/inittab init(8) configuration for BusyBox
#
# Copyright (C) 1999-2004 by Erik Andersen <andersen@codepoet.org>
#
# Note, BusyBox init doesn't support runlevels. The runlevels field is
# completely ignored by BusyBox init. If you want runlevels, use sysvinit.
#
# Format for each entry: <id>:<runlevels>:<action>:<process>
#
# <id>: WARNING: This field has a non-traditional meaning for BusyBox init!
#
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# field is left blank, then the init's stdin/out will be used.
#
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#
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#
# Note: askfirst acts just like respawn, but before running the specified
# process it displays the line "Please press Enter to activate this
# console." and then waits for the user to press enter before starting
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#
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#
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#
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#
# ::sysinit:/etc/init.d/rcS
# ::askfirst:/bin/sh
# ::ctrlaltdel:/sbin/reboot
# ::shutdown:/sbin/swapoff -a
# ::shutdown:/bin/umount -a -r
# ::restart:/sbin/init
# tty2::askfirst:/bin/sh
# tty3::askfirst:/bin/sh
# tty4::askfirst:/bin/sh
#
# Boot-time system configuration/initialization script.
# This is run first except when booting in single-user mode.
#
# ::sysinit:/etc/init.d/rcS
#
# /bin/sh invocations on selected ttys
#
# Note below that we prefix the shell commands with a "-" to indicate to the
# shell that it is supposed to be a login shell. Normally this is handled by
# login, but since we are bypassing login in this case, BusyBox lets you do
# this yourself...
#
# Start an "askfirst" shell on the console (whatever that may be)
# ::askfirst:-/bin/sh
# Start an "askfirst" shell on /dev/tty2-4
# tty2::askfirst:-/bin/sh
# tty3::askfirst:-/bin/sh
# tty4::askfirst:-/bin/sh
#
# /sbin/getty invocations for selected ttys
# tty4::respawn:/sbin/getty 38400 tty5
# tty5::respawn:/sbin/getty 38400 tty6
#
# --More--(86%)
```

```

root@ub2: ~
# /etc/inittab init(8) configuration for BusyBox
# Copyright (C) 1999-2004 by Erik Andersen <andersen@codepoet.org>
#
# Note, BusyBox init doesn't support runlevels. The runlevels field is
# completely ignored by BusyBox init. If you want runlevels, use sysvinit.
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#
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#
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# ::askfirst:/bin/sh
# ::ctrlaltdel:/sbin/reboot
# ::shutdown:/sbin/swapoff -a
# ::shutdown:/bin/umount -a -r
# ::restart:/sbin/init
# tty2::askfirst:/bin/sh
# tty3::askfirst:/bin/sh
# tty4::askfirst:/bin/sh
#
# Boot-time system configuration/initialization script.
# This is run first except when booting in single-user mode.
#
# ::sysinit:/etc/init.d/rcS
#
# /bin/sh invocations on selected ttys
#
# Note below that we prefix the shell commands with a "-" to indicate to the
# shell that it is supposed to be a login shell. Normally this is handled by
# login, but since we are bypassing login in this case, BusyBox lets you do
# this yourself...
#
# Start an "askfirst" shell on the console (whatever that may be)
::askfirst:-/bin/sh
# Start an "askfirst" shell on /dev/tty2-4
tty2::askfirst:-/bin/sh
tty3::askfirst:-/bin/sh
tty4::askfirst:-/bin/sh
#
# /sbin/getty invocations for selected ttys
tty4::respawn:/sbin/getty 38400 tty5
tty5::respawn:/sbin/getty 38400 tty6
inittab

```

```

:shutdown:/sbin/swapoff -a
oot@ub2:~# more inittab
# /etc/inittab init(8) configuration for BusyBox
#
# Copyright (C) 1999-2004 by Erik Andersen <andersen@codepoet.org>
#
# Note, BusyBox init doesn't support runlevels. The runlevels field is
# completely ignored by BusyBox init. If you want runlevels, use sysvinit.
#
# Format for each entry: <id>:<runlevels>:<action>:<process>
#
# <id>: WARNING: This field has a non-traditional meaning for BusyBox init!
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# the specified process to run on. The contents of this field are
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# be unique, although if it isn't you may have strange results. If this
# field is left blank, then the init's stdin/out will be used.
#
# <runlevels>: The runlevels field is completely ignored.
#
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#
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#
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#
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#
# Note: BusyBox init works just fine without an inittab. If no inittab is
# found, it has the following default behavior:
#
# ::sysinit:/etc/init.d/rcS
# ::askfirst:/bin/sh
# ::ctrlaltdel:/sbin/reboot
# ::shutdown:/sbin/swapoff -a
# ::shutdown:/bin/umount -a -r
# ::restart:/sbin/init
# tty2::askfirst:/bin/sh
# tty3::askfirst:/bin/sh
# tty4::askfirst:/bin/sh
#
# Boot-time system configuration/initialization script.
# This is run first except when booting in single-user mode.
#
# ::sysinit:/etc/init.d/rcS
#
# /bin/sh invocations on selected ttys
#
# Note below that we prefix the shell commands with a "-" to indicate to the
# shell that it is supposed to be a login shell. Normally this is handled by

```

```

root@ub2:~#
root@ub2:~#
root@ub2:~# tail inittab
# Example how to put a getty on a modem line.
#::respawn:/sbin/getty 57600 ttyS2

# Stuff to do when restarting the init process
::restart:/sbin/init

# Stuff to do before rebooting
::ctrlaltdel:/sbin/reboot
::shutdown:/bin/umount -a -r
::shutdown:/sbin/swapoff -a
root@ub2:~#

```

(8) 用 find 命令查找命令 useradd 在系统的什么位置 which useradd。

```
root@ub2:~# find / -name useradd 2>/dev/null
/etc/default/useradd
root@ub2:~# which useradd
/usr/sbin/useradd
root@ub2:~#
```

(9) 用 grep 命令查找 etc 目录下含有字符串 “wl0505” 的文件有哪些，并记录 find /etc | grep wl0505 *(查找/etc 目录下所有文件是否包含 wl0505 字符串)

```
root@ub2:~#
root@ub2:~# grep -rl "wl0505" /etc > result.txt
root@ub2:~# cat result.txt
```

(10) 用 date 查看系统时间,把系统时间改为 2007 年 9 月 19 号 14 点 20 分。date 091914202007

```
root@ub2:~# date
Wed Jun  4 20:54:30 +08 2025
root@ub2:~#
root@ub2:~# date 091914202007
Wed Sep 19 14:20:00 +08 2007
root@ub2:~#
```

(11) 用命令 history 查看本用户在本机上进行过的操作。

```
root@ub2:~# history
1  usermod -aG sudo root
2  cd /root
3  mkdir mydir
4  ls
5  apt-get install openjdk-8-jdk
6  apt update
7  y
8  apt-get install openjdk-8-jdk
9  apt install openjdk-8-jdk
10 sudo apt-get update
11 sudo apt-get install software-properties-common
12 sudo add-apt-repository ppa:openjdk-r/ppa
13 apt install openjdk-8-jdk
14 sudo apt-get update
15 apt-get install openjdk-8-jdk
16 sudo add-apt-repository ppa:ubuntu-desktop/ubuntu-make
17 sudo apt-get update
18 sudo apt-get install ubuntu-make
19 umake ide eclipse
20 sudo apt-get install codeblocks
21 sudo nano /etc/apt/sources.list
22 rm /etc/apt/sources.list
23 sudo nano /etc/apt/sources.list
24 apt update
25 sudo apt-get install codeblocks
26 ls
27 ;s
28 cd /root
29 ls
30 cp /etc/inittab mydir/
31 cp /etc/passwd mydir/
32 ls /mydir
33 ls ./mydir
34 cd ./mydir
35 touch mydir/mydoc
```

(12) 将/etc/fstab 文件复制到主目录下,用 ls -l 查看主目录下的 fstab 的详细信息,并记录

```
root@ub2:~# cp /etc/fstab ~
root@ub2:~# ls -l ~/fstab > fstab_info.txt
root@ub2:~#
```

(13) 请用 uname 命令查看系统的信息,包括系统的内核、内核名、主机名。

```
root@ub2:~# uname -a
Linux ub2 4.4.0-142-generic #168~14.04.1-Ubuntu SMP Sat Jan 19 11:26:28 UTC 2019
x86_64 x86_64 x86_64 GNU/Linux
```

```
root@ub2:~# uname -r
4.4.0-142-generic
root@ub2:~# uname -s
Linux
root@ub2:~# uname -n
ub2
```

(14) 请用 free 命令查看当前系统内存的使用情况,并请记录。

```
root@ub2:~# free -h > mem_usage.txt
root@ub2:~# free -h
```

	total	used	free	shared	buffers	cached
Mem:	5.3G	2.0G	3.4G	26M	137M	889M
-/+ buffers/cache:		983M	4.4G			
Swap:	5.5G	0B	5.5G			

(15) 请用 uptime 命令查看系统已运行的时间,并请记录。

```
root@ub2:~# uptime
20:57:56 up 2:07, 2 users, load average: 0.17, 0.09, 0.02
root@ub2:~# uptime > uptime_info.txt
root@ub2:~#
```

(16) 请用 finger 命令查看用户信息或查看默认的用户环境。

```
root@ub2:~#
root@ub2:~# apt install finger
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  finger
0 upgraded, 1 newly installed, 0 to remove and 63 not upgraded.
Need to get 17.3 kB of archives.
After this operation, 68.6 kB of additional disk space will be used.
Get:1 http://mirrors.aliyun.com/ubuntu/trusty/universe finger amd64 0.17-15
.3 kB]
Fetched 17.3 kB in 0s (22.8 kB/s)
Selecting previously unselected package finger.
(Reading database ... 171563 files and directories currently installed.)
Preparing to unpack .../finger_0.17-15_amd64.deb ...
Unpacking finger (0.17-15) ...
Processing triggers for man-db (2.6.7.1-1ubuntu1) ...
Setting up finger (0.17-15) ...
root@ub2:~# finger
```

Login	Name	Tty	Idle	Login Time	Office	Office Phone
r00t	r00t	*:0		Jun 4 18:51 (:0)		
r00t	r00t	pts/0		Jun 4 18:58 (:0)		

```
root@ub2:~#
```

(17) 请用 `hostname` 命令设置系统的主机名为 `osname`。

```
root@ub2:~# hostnamectl set-hostname osname
```

Reboot 后:

```
root@osname:~$
```

五、实验结果及分析

大部分功能都能操作，但现代 Linux 发行版已不再使用 SysVinit 系统，而是采用 `systemd` 作为初始化系统。所以这些系统不存在 `/etc/inittab`。

备注：将代码和实验报告打包发邮件

本实验无代码